

what is package and wt is module in python :

What is a Module?

Imagine a module like a single file (like a page in a book).

It's a Python file (.py) that has some code inside it — like functions, variables, or classes — ready to use.

Example:

Suppose you have a file called `math_tricks.py`:

```
# math_tricks.py
```

```
def add(a, b):
```

```
    return a + b
```

```
def multiply(a, b):
```

```
    return a * b
```

You can use this module in another file:

```
import math_tricks
```

```
print(math_tricks.add(2, 3))  # Output: 5
```

```
print(math_tricks.multiply(4, 5)) # Output: 20
```

What is a Package?

A package is like a folder that holds many modules (many Python files).

A package is a collection of related modules grouped together in one place.

The folder must have a special file called `__init__.py` (even if it's empty!) to tell Python, "Hey, this is a package!"

Example:

```
math_fun/      <- this is the package (a folder)
__init__.py    <- (this tells Python it's a package)
add_numbers.py <- (a module)
multiply_numbers.py <- (another module)
```

You can use it like this:

```
from math_fun import add_numbers
from math_fun import multiply_numbers

print(add_numbers.add(2, 3))
print(multiply_numbers.multiply(4, 5))
```

Note:

Module A single page or story

Package A full book (with many pages/stories)

Module = a Python file with code.

Package = a folder with many Python files inside.

python regular expression (regex):

In Python, you use the re module to work with regular expressions.

A regular expression is a sequence of characters that forms a search pattern, mainly used for string searching and manipulation.

1. Basic Syntax

First, you import the re module:

```
import re
```

2. Common Functions in re module

Function	Purpose
re.match()	Check for a match only at the beginning of the string
re.search()	Search for a match anywhere in the string

`re.findall()` Find all matches in the string
`re.sub()` Replace matches with another string
`re.split()` Split the string by the matches

3. Simple Examples

`re.match()`

```
import re
```

```
result = re.match(r"hello", "hello world")
```

```
if result:
```

```
    print("Matched!")
```

`re.search()`

```
import re
```

```
result = re.search(r"world", "hello world")
```

```
if result:
```

```
    print("Found!")
```

`re.findall()`

```
import re
```

```
words = re.findall(r"\w+", "Hello, how are you?")
```

```
print(words) # ['Hello', 'how', 'are', 'you']
```

`re.sub()`

```
import re
```

```
text = re.sub(r"\s+", "-", "Hello how are you")
```

```
print(text) # Hello-how-are-you
```

```
re.split()
```

```
import re
```

```
parts = re.split(r"\s+", "Hello how are you")
```

```
print(parts) # ['Hello', 'how', 'are', 'you']
```

4. Special Characters in Regex

Symbol Meaning

. Any character except newline

^ Start of string

\$ End of string

* 0 or more repetitions

+ 1 or more repetitions

? 0 or 1 repetition

{n} Exactly n repetitions

{n,m} Between n and m repetitions

[] Set of characters

、 、

() Grouping

\d Digit

\D Non-digit

\s Whitespace

\S Non-whitespace

`\w` Word character (letters, digits, _)

`\W` Non-word character

5. Example: Email Validation

```
import re
```

```
email = "test@example.com"
```

```
pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
```

```
if re.match(pattern, email):
```

```
    print("Valid Email")
```

```
else:
```

```
    print("Invalid Email")
```